Categorizing Consumer Grievances in Buy Now, Pay Later Services: An LDA Analysis of CFPB Complaint Narratives

AP Research

4996 words

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This study uses a text mining tool called Latent Dirichlet Allocation (LDA) to find themes in over 5,000 complaints submitted to the Consumer Financial Protection Bureau about Buy Now, Pay Later services. While the current academic literature is focused on the general consequences of BNPL to the economy using credit card data and theoretical analyses, this research focuses on the perspective of consumers about BNPL. The LDA model derived 12 different themes, which showed that BNPL consumers were complaining about legal/structural issues, issues with fraud, and new technological issues like app functionality and the return process with BNPL. This indicates that what BNPL consumers are complaining about may not be entirely in line with what the academic literature is writing about.

Introduction

Financial technology (FINTECH) companies can be defined as companies that utilize technology to enhance financial services (Di Maggio & Yao, 2018). Fintech companies have grown considerably in recent years, becoming a major part of the credit sector. This expansion is evident in services such as Buy Now, Pay Later (BNPL), whose reach has expanded rapidly, with 20% of the 200 monthly rotating respondents from the New York Federal Reserve Bank claiming to have used BNPL (Maureengan, 2024). BNPL is defined as short-term loans that allow consumers to buy goods or services and pay off the balance over a fixed number of installments over a set period of time, interest-free (Hayashi & Routh, 2018). A problem that's been identified with BNPL is the lack of regulation surrounding it (Johnson et al., 2021). This can create

concerns for consumer safety as BNPL can increase consumer spending through deceptive advertising, hidden fees, and overdraft charges (Cook et al., 2023; deHaan et al., 2024). Furthermore, BNPL services with low interest rates have been found to attract consumers who are single, younger, and on lower incomes, groups often identified as being financially vulnerable consumers (Chen et al., 2024). This means that consumers who are already financially vulnerable may be prone to overspending due to BNPL. The Consumer Financial Protection Bureau (CFPB) is the regulatory authority in charge of regulating BNPL. However, existing BNPL regulations are no longer being enforced (*CFPB Announcement Regarding Enforcement Actions Related to Buy Now, Pay Later Loans*, 2025). This leads to further concerns that BNPL's lack of oversight will allow it to continue negatively impacting lower-income consumers who are considered more vulnerable.

Perspectives on BNPL

There are a variety of perspectives on the benefits vs risks of BNPL in the literature, but they form a consensus that further regulation of BNPL is necessary to protect consumers. What they disagree on, however, is the degree of regulation needed: balancing the benefits of innovations against the consumer risks they may pose. Research from five different experimental studies found that the innovations from BNPL significantly increase consumer spending, greatly benefiting retailers (Li et al., 2025). However, the researchers note that these corporate benefits should not come at the expense of consumers, especially financially vulnerable ones who may be harmed by frequent overspending. This view is supported by other researchers, who believe that BNPL has several innovative benefits to consumers and businesses but can create new consumer risks of financial overextension, particularly among the most vulnerable consumers (Mishra et

al., 2025; Di Maggio & Yao, 2018). The most important of these innovations is increased credit access for consumers who would have otherwise struggled to obtain it from a bank. This reflects BNPL's broader business model. Fintech companies such as BNPL services first expanded their market share by loaning money out to higher-risk consumers whom banks would not loan to. While these companies have shifted to lower-risk consumers in recent years, they are less likely to reject consumers on risky behavioral traits (Di Maggio & Yao, 2018). This means that BNPL companies are lending out money to financially vulnerable consumers who would not be able to get credit with traditional banking institutions. Donou-Adounsi and Leslie-Piper (2025) found with national survey data that BNPL can increase the spending of more vulnerable consumers. However, they noted that BNPL is both a behavioral amplifier of household financial risk and a credit access tool, meaning the reason that vulnerable consumers are spending more is that they have access to more credit. So, a concern may be that over-regulating BNPL would lead to less credit available for vulnerable consumers since BNPL companies are more willing than traditional banks to lend out money to vulnerable consumers (and often without interest rates).

An alternate perspective among academics is that BNPL regulations should be harsher due to the overspending effect of BNPL on consumers. The most common argument from these researchers is that BNPL regulations are needed since currently there are little to no regulations. They claim that regulations are needed to protect consumers since understanding the increasing complexity of Fintech tools, such as BNPL, is beyond the responsibilities and capabilities of the average consumer (Johnson et al., 2021). Under this viewpoint, BNPL services take advantage of the lax regulatory situation to manipulate vulnerable consumers into taking on more debt and increasing their spending. An analysis of different experimental studies by Ashby et al. (2025) found that BNPL services break up the price of goods into 4 installment prices, making the first

installment price the cheapest. This implicitly makes consumers think that the good is cheaper than it really is through a concept known as mental accounting. This behavioral economics principle states that consumers will accept losses more readily if they are broken up into parts, such as the four installment payments (Thaler, 1985). When consumers use BNPL, they think they are spending less money than they really are because it is broken up into smaller installments. This can cause overspending among consumers, specifically those who exhibit risky behavioral traits, which is the origin of some of the academics' perspectives on the service. Contrary to the academics who view BNPL more positively, these academics see the issue of BNPL regulation as more urgent and believe that the weak regulatory environment that exists today is a breeding ground for BNPL services to take advantage of consumers who don't know any better, using mental accounting to their advantage.

Study Justification

Despite the exhaustive body of literature on the economic effects of BNPL services on vulnerable consumers, there are limitations to these works. Crucially, these studies all focus on the general economic effects of BNPL on vulnerable consumers using credit data, simulation studies, or applying general frameworks to the BNPL market. While this provides an excellent context of the problems with BNPL, it leaves vulnerable consumers out of the discussion. While there are studies that use survey data, they are quantitative analyses of those surveys, which leave out consumers' voices. This is important because vulnerable consumers deserve to have their voices heard on the BNPL issue, which directly affects them. The current field of BNPL is debating the regulation of BNPL services without considering the perspectives of consumers.

One of the places consumers often share their perspective is the CFPB consumer complaints survey; for this reason, this study utilizes a method named Latent Dirichlet Allocation

(LDA) to analyse consumer complaints. Bastani et al. (2019) used LDA on all of the consumer complaint narratives to find the most common words in these complaints and form them into similar topics, which could indicate problems these consumers had. The purpose of that research was to demonstrate the effectiveness of LDA in consumer-oriented or regulatory research. This study applies this framework of using LDA on CFPB consumer complaints, but with just BNPL complaints, in order to indicate which issues BNPL consumers are complaining about. This method of LDA is being used to capture consumer perspectives on BNPL, which will, in turn, answer the research question: "How are Buy Now, Pay Later Consumer Complaints from the Consumer Financial Protection Bureau indicative of the scholarly discussion surrounding Buy Now, Pay Later Services?"

Methods

Latent Dirichlet Allocation (LDA)

Latent Dirichlet Allocation (LDA) is an unsupervised machine learning text mining technique that groups the text in documents into similar "topics," which are essentially the themes of those words (Blei et al., 2003). In this study, LDA was used to find latent topics in over 5,000 consumer complaints in order to identify consumers' concerns regarding BNPL. This method was first developed by Blei et al. in 2003, but much of this paper's method is based on Bastani et al.'s framework, which also used LDA to analyze consumer complaints (Bastani et al., 2019). LDA was chosen over qualitative methods of categorizing text into themes because once the LDA process is set up, the words are automatically categorized into topics by the model. This allows the user to analyze considerably larger datasets compared to manual qualitative analysis, with this study's dataset being over 5,000 documents. This increases the generalizability of this study since it analyzes a large number of consumer complaints from across the United States.

Additionally, using LDA may be less biased than creating topics or themes qualitatively, since it is an automated process that does not have as many opportunities for human bias to interfere with the results. When comparing different methods of topic modeling, or the quantitative analysis of text, LDA was shown to be the most superior method in terms of accuracy, which is typically measured using *perplexity scores* and *coherence scores*. A perplexity score represents how well the LDA model predicts new data, with a lower perplexity score being ideal (Bastani et al., 2019; Wahid et al., 2022). A coherence score measures the semantic similarity between the most frequent keywords in a topic, with a higher coherence score being ideal. In this case, LDA had the lowest average perplexity scores and highest coherence scores even as the number of documents greatly increased, indicating a highly accurate method (Blei et al., 2003).

Data Collection

The dataset being used is from the CFPB consumer complaint survey (*Search the Consumer Complaint Database | Consumer Financial Protection Bureau*, 2026). This database includes all of the CFPB consumer complaints, so for this study, it was filtered by narrative, year, and company before being used for LDA. This could be easily done on the CFPB's website using an array of filters on the sidebar. Not all CFPB complaints contain text or the consumer's permission to display their complaint, which required filtering by narrative in order to obtain text that the LDA model could analyze. The range of complaints was filtered from the beginning of 2021 to August 2025, because complaints after 2021 would ensure that the Covid pandemic would not significantly skew the data, and August 2025 was chosen to ensure that the government shutdown would not interfere with the results. The dataset was originally filtered to include only the 6 largest BNPL companies: PayPal, Klarna, Affirm, Afterpay, Zip, and Sezzle, because together they account for over 94 percent of BNPL lending (Schulz, 2025). However,

two of the companies, PayPal and Afterpay, were later excluded to ensure the validity of the data. Paypal needed to be excluded because it is primarily a payment platform, not a BNPL platform; therefore, Paypal was removed in order to avoid erroneously including non-BNPL complaints in the data. Afterpay was excluded because it contained fewer than 100 CFPB complaint narratives, which is under the threshold for the typical sample sizes required for accurate use of LDA (Chen et al., 2024).

The Orange Software

Overall, this study analyzed over 5,000 complaints from the CFPB database, but the exact number of participants and their information was unknown due to CFPB confidentiality policies. After finishing the data filtering, the complaint narratives were exported into an online open source data mining site named "Orange," which can run LDA using several different data analysis widgets instead of code (Ljubljana, 2024). The creation of parameters for filtering the dataset and creating the list of stop words was conducted manually, and then the Orange software was able to automatically preprocess the data and run it through the LDA model. This part of the procedure is explained in detail below.

LDA Procedure and Preprocessing

Before the LDA algorithm was applied, the dataset was converted into a format that the LDA model could calculate and interpret. Bastani et al. (2019) simplified this process into five steps: converting the text into lowercase, removing special characters, removing stop words (common words that have little semantic value), stemming (reducing words to their stem), and creating a term document matrix. After ensuring that the correct widgets were connected with the correct settings, the Orange site was able to run every pre-processing step on its own except for the step to remove stop words. This required manually creating a document of stop words, then

adding this document to the Orange pre-processing widget. The selection criteria of stop words were based on a pre-processing framework for LDA developed by Zimmerman et al. (2024), known as the Zimm approach. In this framework, an iterative process known as Coherent Utility Processing (CUP) is utilized to remove stop words. This technique involves creating a word cloud based on the words in the unfiltered document, then identifying and removing both generic and domain-specific stop words. Generic stop words are words that are commonly used (such as "like" or "as"), but are not relevant to the scope of this research, since they do not contain content. Domain-specific stop words are words that are likely to show up in the CFPB complaints, which, while specific to the topic being researched, do not provide any valuable information. For example, although the acronym "BNPL" is relevant to the scope of the research, it does not provide specific insight, since all the complaints in question would be about BNPL anyway. After removing the stop words, the word cloud was continuously run until no stop words could be observed. The difference between this study's word cloud pre-CUP and post-CUP is shown below.

Figure 1

Word Cloud of the filtered CFPB dataset before (above) and after (below) pre-processing





Note: This graph was made with the Orange software. The weight refers to the relative frequency of each word in the dataset, and this corresponds to the size of each word. The colors are purely for visual purposes.

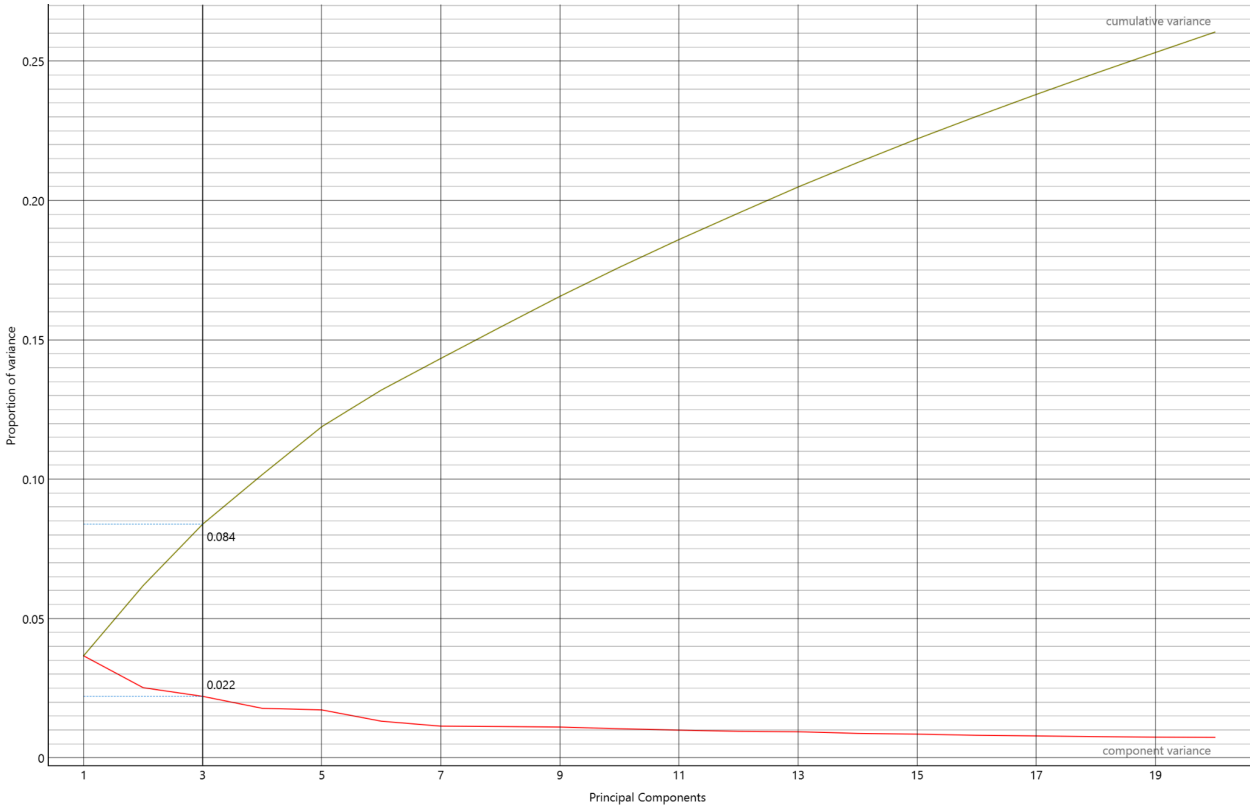
Identifying the Correct Number of Topics (K Number)

After pre-processing the data, the next step was to identify how many topics the model should find, known as the *k number*. Because the LDA algorithm assumes that the number of topics is known and fixed, this number must be chosen carefully. If the k number is too small, the topics will be broad and of little value. If the k number is too large, the themes will be oversaturated and redundant (Amjad et al., 2023). Most LDA researchers have found trial and error to be the most effective way to find the k number; they would run the model continuously with different k numbers until they find the one with the best perplexity scores and coherence scores (Amjad et al., 2023; Bastani et al., 2019; Blei, 2012; Blei et al., 2003). Unlike this previous research, this study utilizes the Zimm approach, which uses a principal component analysis, a type of analysis that objectively determines the k value (Zimmermann et al., 2024). The k number was found by looking at the area of the graph where the red line that represents component variance (which

shows how much unique information each topic has) begins to flatline around 12 topics. Thus, the ideal k number is around 12, and including any more topics would yield diminishing returns for the results.

Figure 2

Changes in the proportion of cumulative and component variance as a function of principal components (number of topics)



Note: The x-axis measures "principal components," which in this case are the number of topics selected, while the y-axis measures "proportion of variance," which is the percentage of the total dataset's information that each topic would capture. The green line (labeled cumulative variance) shows the total sum of the information gathered in the dataset. The red line (called component variance) shows how much unique information each topic has.

Validity

The credibility of the results of the LDA algorithm is usually determined by its perplexity and coherence scores. The final perplexity and coherence scores were 76.82 and 0.47, respectively. LDA researchers advise against using perplexity and coherence scores as absolute metrics, because each LDA model might have a different range of scores. Instead, they recommend using these scores as a relative comparison between pre-processing and post-processing (Chang et al., 2009). In the case of this research, the original scores were 80.50 and 0.31, respectively, representing an improvement. While perplexity and coherence scores are used to quantitatively describe the analysis's credibility, there is an aspect in which the researchers must qualitatively label the topics that the LDA model generates. For the qualitative component, another form of validation is needed, most frequently an inter-rater reliability test or its equivalent (Bastani et al., 2019; Cook et al., 2023; Wahid et al., 2022). In this study, an individual with proficient knowledge in BNPL acted as a peer reviewer in an inter-rater reliability test. They were given the keywords grouped by topic, but with a randomized topic order and no prior knowledge of the topics or assistance from the researchers. Then, they were given a list of the qualitative topic labels created by the author of this paper and were asked to sort them. For example, if given the series of words, “USC, Section, State, Right, Privacy,” the rater would match it with Topic 1, “Violation of Federal Consumer Rights.” The analysis of the results is given in the section below.

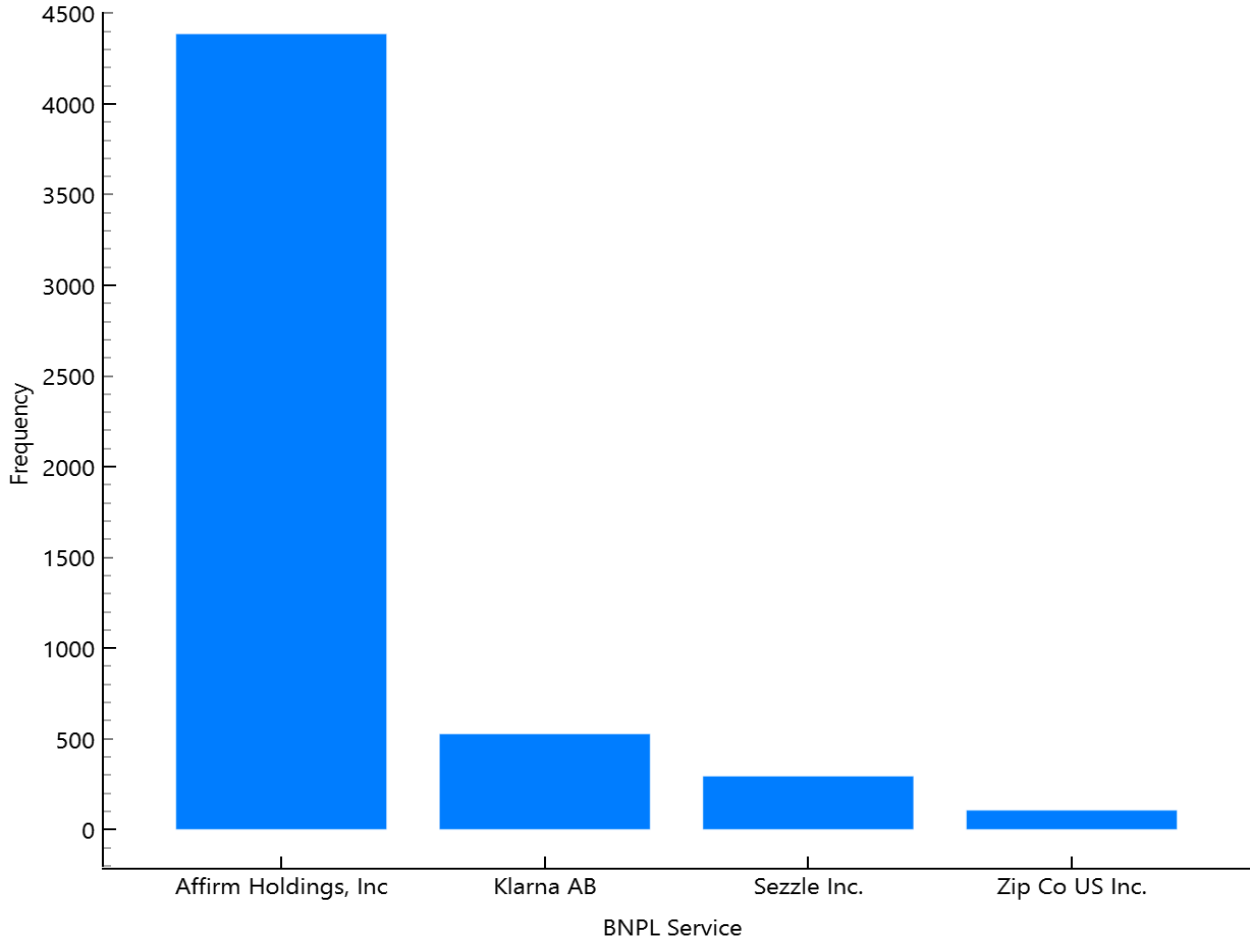
Results

Complaint Distribution

The LDA model was able to yield coherent topics from the CFPB consumer complaint database. The sample size was 5,313 narrative complaints, which were analyzed by LDA, resulting in 12 distinct topics about BNPL consumer complaints. The distribution of these complaints by BNPL service is shown in Figure 3.

Figure 3

Frequency of CFPB complaints by BNPL provider (N=5,31



In Figure 3, Affirm Holdings has the most consumer complaints by a substantial margin, with about 82% of all analyzed complaints, followed by Klarna AB, Sezzle Inc, and Zip Co US Inc. Despite Affirm's overwhelming majority, its prevalence in the dataset did not interfere with the LDA process or results. Because LDA relies on probability for its calculations, the results are based on relative frequency and not absolute frequency. Therefore, Affirm having a higher number of complaints does not skew the results in a disqualifying manner.

LDA Topics

To ensure the validity of the human-labeled topics, an Inter-Rater Reliability test was conducted and yielded a score of 83% agreement using the simple percent agreement formula. The 12 topics, their keywords (in order of relevance left to right), and their validated labels are shown in Table 1 below.

Table 1:

LDA-Derived Topics with identifying keywords and qualitative labels

Topics	Key Words (in order of relevance)	Label
Topic 1	Usc, section, right, state, privacy, violated, agency, creditor, purpose, written	Violation of federal consumer rights
Topic 2	Balance, late, debt, report, account, full, credit, reported, showing, collection	Consumer issues with BNPL credit balances
Topic 3	Debt, violation, law, identity, report, theft, collection, consent, federal, legal	Legal Disputes regarding identity theft
Topic 4	Refund, item, order, returned, return, merchant, product, purchase, refunded, purchased	Issues regarding refunds and returns
Topic 5	Refund, merchant, dispute, despite, transaction, complaint, support, purchase, resolution, financial	Issues regarding customer service
Topic 6	Inaccurate, fcra, debt, report, credit, account, reported, late, accurate, fair	Reported violations of the fcra
Topic 7	Interest, financial, negative, already, continue, caused, resolved, concern, paying credit	Unresolved Payment or credit issues
Topic 8	Card, purchase, bank, app, use, customer, make, credit, account, reason	Issues with app/process functionality
Topic 9	Case, please, asking, regarding, know, report, able, situation, personal, yet	Consumer pleas and asking for assistance
Topic 10	Call, email, get, got, phone, saying, nothing, purchase, anything, account	Issues with calling customer service
Topic 11	Identity, theft, opened, fraudulent, account, name, report, fraud, credit, open	Identify theft and other issues with fraud
Topic 12	Phone, keep, tried, account, access, email, help, past, getting, get	Issues with account access

While each topic is distinct, these topics can be grouped by the general issue of BNPL that they describe. One of these groups, which included Topics 5, 9, and 10, was omitted from the analysis, since they describe common customer service issues that could be expected across all CFPB complaints, regardless of sector. The rest of the topics can be sorted into 3 groups: regulatory and legal disputes, fraud vulnerabilities, and digital platform functionality. The largest group of topics is the one about regulatory and legal disputes, which includes Topics 1,2,6, and 7. Additionally, topics 3 and 11 relate to fraud vulnerabilities, while topics 4, 8, and 12 relate to digital platform functionality. While Topic 3, “legal disputes regarding identity theft,” could have fallen into either legal disputes or fraud vulnerabilities, it was placed into fraud vulnerabilities

due to keywords such as “theft” and “identity,” which indicated that the central issue of this topic was fraud, and not legality.

Statistical Variance Across BNPL Services

In order to determine whether the prevalence of these consumer issues varied by BNPL service, a one-way analysis of variance (ANOVA) was run. The independent variable was the BNPL provider, while the probability that a BNPL service would fall into a topic was the continuous dependent variable. For example, the results from Topic 1 “Violation of federal consumer rights” can be observed in Figure 4.

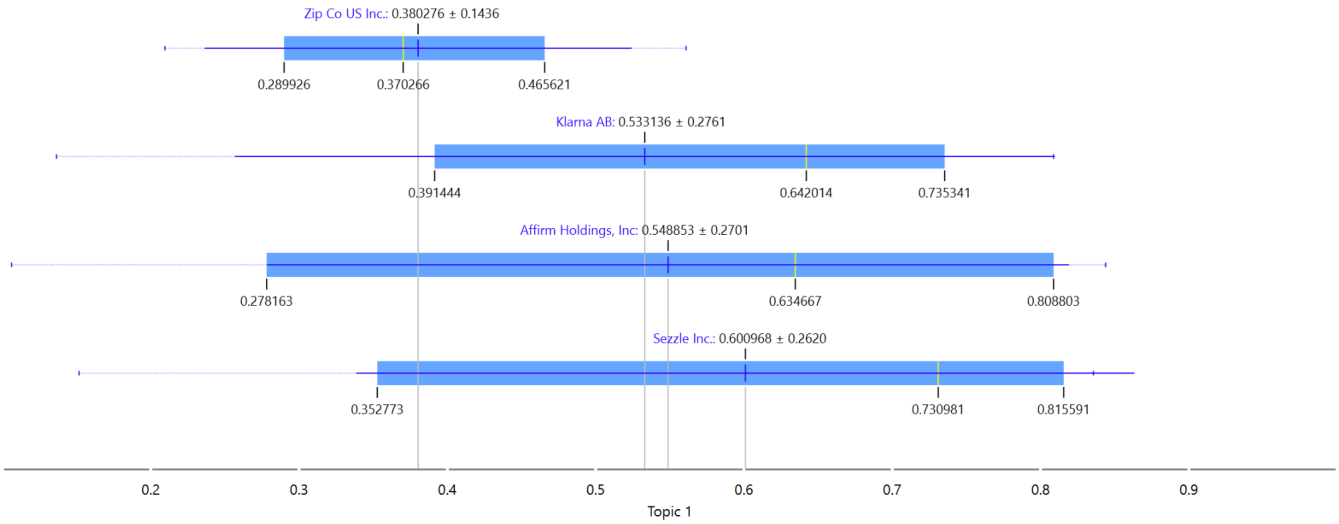


Figure 3: BNPL Service by topic relevance (ANOVA 7.123/P<0.001)

Box plot of BNPL Service’s measured by LDA topic probability score for topic 1

There is a significant effect of BNPL service on its probability to appear in topic 1, indicating that CFPB complaints are not uniform across all BNPL services ($F(3,5309)=2.61$, $p<0.001$). Similar results were observed across topics, revealing that there is significant variance between BNPL services when it comes to each topic.

Discussion

Relationship Between BNPL Scholarly Research and LDA Topics derived from CFPB complaints

The purpose of this study was to compare the themes from BNPL consumer complaints from the CFPB to the issues identified by the BNPL scholarly literature. Some of the LDA topics were in agreement with the scholarly literature, while others identified new issues less discussed by those researchers. For example, Topic 1 from the "Regulatory and Legal Disputes" section was given the label "Violation of Federal Consumer Rights" due to key words in that topic, such as "USC" (which stands for United States Section Code), "right," "state," and "violated", which suggests that consumers are complaining about violations of their rights. This is similar to what was argued by Johnson et al., who argued that violations of federal consumer rights might be a byproduct of the lack of regulation regarding BNPL (Johnson et al., 2021). While these issues are not always directly discussed by the academic literature, they point to the same problem of regulatory arbitrage, which is the BNPL services avoiding federal regulation.

On the other hand, there were several topics that diverged from the scholarly literature, such as Topic 4, which is labeled as "Issues with refunds and returns." There has been little focus on refunds and returns in the BNPL literature, but the prevalence of customer complaints on this topic suggests that it may be an issue of high importance. Some potential reasons that refunds and returns could be further complicated in the BNPL sector are the nature of split payments and that the customer receives the goods before they pay in full. The CFPB had tried to standardize the refund process with an interpretative ruling, but it had since distanced itself from that ruling due to new regulatory shifts (Krause, 2025). Similarly, Topic 8 is labeled as "Issues with app/process functionality." This relates to the current scholarly view that BNPL is designed to have reduced functionality, which could harm consumers if unregulated (Ficawoyi Donou-Adonsou & Neleen Leslie-Piper, 2025; Hayashi & Routh, 2018). However, Topic 8 takes

this idea further by suggesting that this phenomenon is being observed increasingly in the technological aspects of BNPL, such as its apps.

Overall, the CFPB complaints identify and expand upon several of the same problems that the BNPL research does. This can be seen in some of the topics that were found by the LDA model, such as those in the category "Legal and Regulatory Complaints." However, the emergence of several new topics of concern suggests that there is some discrepancy between the issues outlined in the literature and those found in the consumer complaints. A major reason for this could be that vulnerable consumers who are filing CFPB consumer complaints about BNPL do not know the terminology that is commonly used by researchers. This is a similar problem that Bastani et al. ran into during a similar study (Bastani et al., 2019). However, since these discrepancies have been observed on a broad thematic level through groups of topics as opposed to individual words, this explanation appears unlikely. If these complaint topics are representative of consumer complaints, LDA could instead be a useful tool in identifying gaps in academic research worthy of exploration, particularly based on consumer input.

Relationship between BNPL topics and overall CFPB topics

In addition to the well-documented and recent issues concerning BNPL, some other patterns emerged that are symptomatic of the overall consumer credit industry. One such topic was Topic 11, which was labeled as "identity theft and other issues of fraud." Similar topics about fraud were identified by Bastani et al. 2019 when they used LDA on the entire CFPB dataset, showing continuity with the issue of identity theft and fraud from the consumer credit sector to BNPL. This shows that utilizing LDA to analyze consumer complaints could be used to back up wider sector concerns around identity theft, with similar BNPL-specific consumer concerns.

Distribution of complaints by BNPL service

While all of the patterns above exist in the BNPL consumer complaints, this does not mean that these topics are equally distributed among BNPL services. In fact, the ANOVA test found that a high degree of variance in the dataset could be accounted for by the type of BNPL service used. This means that some BNPL services are structured in such a way that could lead to different consumer complaints per service. For example, Sezzle had the highest mean in Figure 3, which means that they are structured in a way that causes more consumers to complain about violations of federal consumer rights (which is the label for Topic 1). In terms of overall size, Affirm had the largest amount of complaints by far (over 80%), which could potentially mean that Affirm's practices cause more overall complaints to be had.

Conclusion

In conclusion, the guiding question examined in this research was, “How are LDA topics derived from CFPB complaints about BNPL indicative of the scholarly discussion surrounding BNPL?”. Four main observations shed light on this question of inquiry. The first is that consumers are complaining about legal issues, such as violations of their rights and structural BNPL issues. These issues are discussed in the academic literature, specifically in relation to BNPL services avoiding federal regulation. The second pattern is that BNPL has the same fraud vulnerabilities already present in consumer credit. While the academic literature discusses these vulnerabilities, having the additional view from the BNPL consumers’ perspective may be important in order to address this issue. The third pattern is that consumers specifically need to deal with new online-based problems of BNPL that researchers have not yet explored. This was shown in topics four and eight, which discussed app functionality and returns. This problem is not as covered by the current literature, but this gap is crucial since it will become more relevant

in the future as financial services become more technologically immersed. The fourth pattern is that not all BNPL services are equally responsible for each type of complaint. For instance, the distribution of each type of complaint differed across companies. Additionally, Affirm had the most complaints, which could suggest differences in the way Affirm runs its service that may be more harmful for consumers compared to the other BNPL firms.

Implications

Overall, these topics show that BNPL consumers may face many of the same challenges that they have been facing in previous forms. This was shown specifically with Topic 11 (identity theft and other issues of fraud). If consumers are still complaining about identity theft and fraud, it may show that problems surrounding older forms of consumer credit remain with BNPL. The current academic research on BNPL suggests that many of these issues could be caused by a lack of BNPL regulation (Johnson et al., 2021; Mishra et al., 2025). If BNPL services are not properly incentivized to solve these issues, it may fall on future regulators to create new regulations that properly aim to solve this problem.

Unlike other forms of traditional credit, BNPL's structure is entirely reliant on technology. This can give way to new forms of problems consumers may face, which have been unknown until recently to the research and regulatory field. An example of this is Topic 4 (issues with refunds and returns) and Topic 8 (issues with app process/functionality). Both highlight new aspects of BNPL that cause enough distress for consumers that they feel the need to submit formal complaints. While these topics may seem trivial, especially vulnerable consumers can be harmed by these inefficiencies. If a vulnerable consumer has difficulty navigating the BNPL website or refunding an item they bought, it may push them closer to financial ruin. While the positive aspects of BNPL innovation (such as its inclusive lending) are often discussed in the

academic literature, regulation is becoming a popular stance among researchers. Chiu (2025) presents the idea that regulation must be continuously updated in order to cover innovations and prevent regulatory arbitrage. BNPL is no different; these complaint topics suggest that regulation must be updated to cover all aspects of BNPL, including its technological design. However, seeing that BNPL services do not have the correct incentives to fix these problems and regulators are not keeping up with technological innovation, it seems like the current regulatory arbitrage is bound to continue and worsen, leading to further CFPB complaints. This will all have an adverse impact on BNPL consumers (especially vulnerable ones) who will need to figure out how to deal with these new problems on their own.

Unlike previous research on BNPL, this paper used LDA to research consumer complaints about BNPL, filling in that particular gap. LDA was effective in producing coherent topics pertaining to BNPL, showcasing its usage for others researching BNPL or similar fields. Researchers and regulators could use LDA to identify which aspects of BNPL are most in need of regulation. Allowing regulators to target regulations on the largest issues with BNPL instead of the service as a whole could help ensure that BNPL does not lose its innovative benefits. Additionally, this builds off of Bastani et al. 's work using LDA for other CFPB complaints and further shows that LDA can be used for other aspects of finance research to identify themes in complaints.

Limitations

While the LDA-derived themes showed that consumers were concerned about both ongoing and new technical issues with BNPL, there were some limitations of this study. Interestingly, a vast majority of consumer complaints were from the BNPL service Affirm (over 80 percent). This did not disturb the information regarding which BNPL services were most

representative of each topic because that information is based on probability means, which are not impacted by total frequency. However, this could have caused certain complaints to be overrepresented in the overall dataset, causing the perspective of Affirm users to possibly be more represented in the themes. Additionally, the themes that LDA derives are more general rather than specific because it contextually groups large amounts of text, meaning the labels are bound to be general. This means that the findings of this paper do not point to specific problems BNPL consumers are complaining about (like a specific feature or payment option) but rather general problems (like identity theft or confusing websites). Since this data was derived from the CFPB consumer complaints data, it is representative of those who submitted CFPB complaints, not the general BNPL consumer population. Therefore, certain consumers, less likely to submit complaints, like more vulnerable consumers, may have been less represented in these findings.

Directions for Future Research

Future research could focus on using LDA to look at other types of data, like those from credit unions or banks. This would provide results that are potentially more representative of the general population. Additionally, researchers could use LDA to analyze social media posts about BNPL since this more informal setting may make it easier to represent consumers' viewpoints. Finally, researchers could use a case study to measure the effectiveness of BNPL regulation in a specific state. This would move BNPL research beyond identifying problems and into identifying some of the ways this regulatory arbitrage could be halted.

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